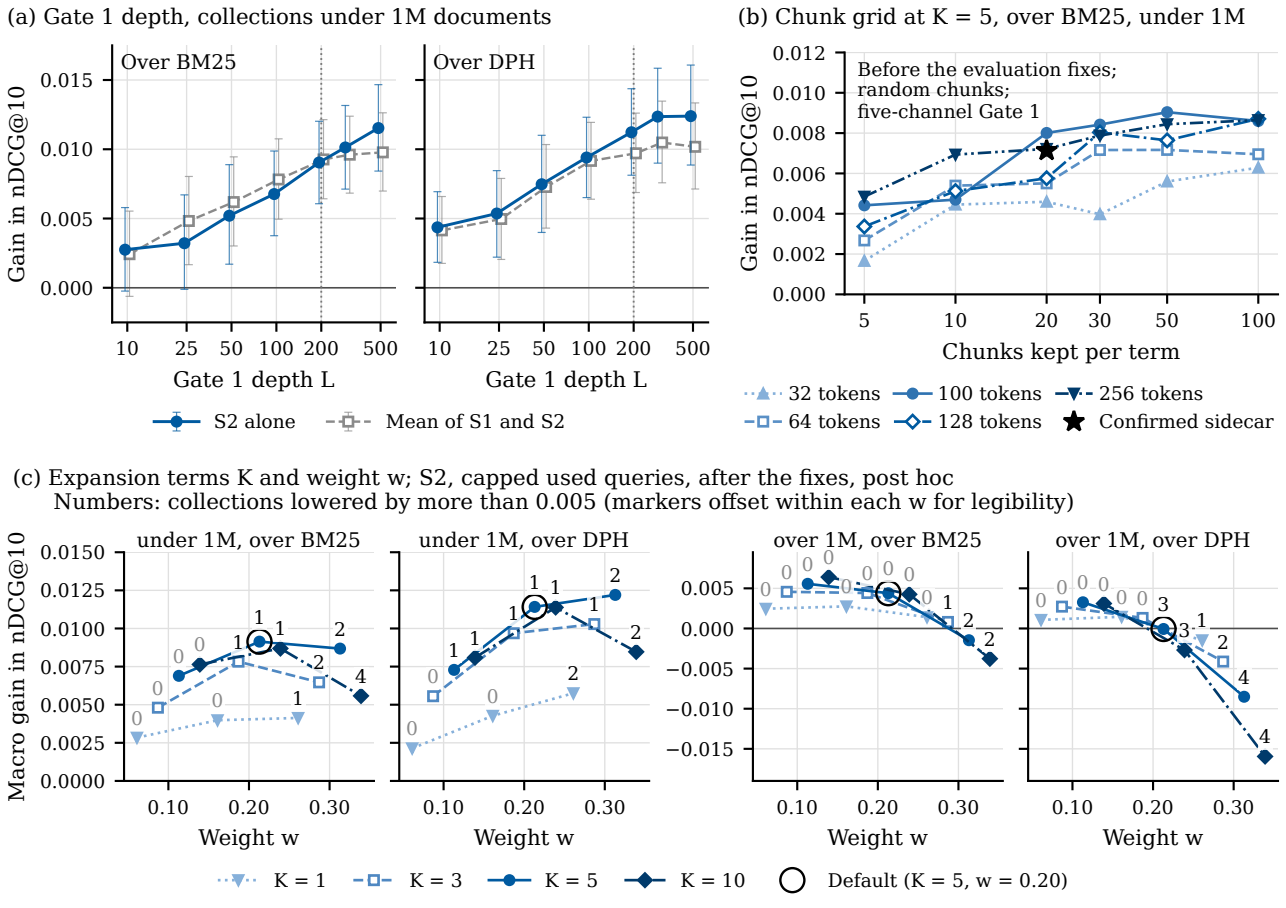
(a) Gate 1 depth, collections under 1M documents
Over BM25
Over DPH
Gain in nDCG@10
0.015
0.010
0.005
0.000
10
25
50
100
200
500
Gate 1 depth L
S2 alone
Mean of S1 and S2
(b) Chunk grid at K = 5, over BM25, under 1M
Before the evaluation fixes;
random chunks;
five-channel Gate 1
0.012
0.010
0.008
0.006
0.004
0.002
0.000
5
10
20
30
50
100
Chunks kept per term
32 tokens
64 tokens
100 tokens
128 tokens
256 tokens
Confirmed sidecar
(c) Expansion terms K and weight w; S2, capped used queries, after the fixes, post hoc
Numbers: collections lowered by more than 0.005 (markers offset within each w for legibility)
under 1M, over BM25
under 1M, over DPH
over 1M, over BM25
over 1M, over DPH
Macro gain in nDCG@10
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0.005
0.000
−0.005
−0.010
−0.015
0.10
0.20
0.30
Weight w
K = 1
K = 3
K = 5
K = 10
Default (K = 5, w = 0.20)